

# Styled Components - Complete Guide

## ■ Basics of Styled Components

Styled Components is a CSS-in-JS library for React. It allows writing CSS directly inside JavaScript, scoped to components, making styles reusable and maintainable.

Example:

```
import styled from 'styled-components'; const Button = styled.button` background: #4caf50; color: white; padding: 10px 20px; border-radius: 8px; `; export default function App() { return Click Me; }
```

## ■ Using ThemeProvider

ThemeProvider allows defining global design tokens like colors and breakpoints.

Example:

```
import { ThemeProvider } from 'styled-components'; const theme = { colors: { primary: '#4caf50', danger: '#f44336', }, }; const Button = styled.button` background: ${props => props.theme.colors.primary}; color: white; padding: 10px 20px; `; export default function App() { return ( Save ); }
```

## ■ Media Queries with Styled Components

```
const Box = styled.div` width: 400px; height: 200px; background: teal; @media (max-width: 768px) { width: 200px; background: tomato; } `;
```

## ■ Styled Components Interview Q&A;

- **Q: What are Styled Components?**  
A: A library for styling React components using tagged template literals and CSS-in-JS methodology.
- **Q: Difference between Styled Components and CSS modules?**  
A: Styled Components co-locate styles with components and support dynamic styling via props. CSS modules scope CSS locally but remain separate files.
- **Q: How does theming work in Styled Components?**  
A: Using ThemeProvider, you can pass a theme object and access its properties via props inside styled components.
- **Q: How to handle responsive design?**  
A: Use media queries directly inside styled components or define centralized breakpoints in theme.
- **Q: Pros of Styled Components?**  
A: Scoped styles (no clashes), dynamic styling, theming support, and better component reusability.
- **Q: Cons of Styled Components?**  
A: Runtime overhead, larger bundle size compared to plain CSS, and learning curve for new developers.

- **Q: How to extend styles in Styled Components?**

A: Use `styled(SomeComponent)`...`` to create a new styled component based on existing styles.